

capital-intensive, battery swapping stations offer a quick turnaround time, making them particularly suitable for scenarios requiring rapid charging.

Battery swapping's suitability is most pronounced for two- and three-wheelers, where its adoption has gained traction, particularly in India. Shazan says, "Startups like Bounce have pioneered battery swapping for electric scooter fleets. However, the technology's dominance is concentrated in China, with over 2,200 swap stations for cars alone established by the end of 2022. Nio, a major Chinese automotive manufacturer, leads this movement and has even expanded its technology to Europe, albeit on a smaller scale."

While primarily prevalent in China, battery swapping's influence is extending beyond its borders. It has gained traction for commercial vehicles, notably taxis and micro EVs. However, its suitability for passenger cars, especially in emerging markets like India, where private EV adoption is nascent, remains uncertain.

Battery swapping's success relies on government backing, operational efficiency, and market demands. As the passenger EV landscape evolves and gains momentum, battery swapping could emerge as a viable option, particularly in scenarios demanding rapid turnarounds, such as commercial ride-hailing. As this dynamic industry evolves, trends are evident, yet the course of battery swapping's adoption hinges on technological advancements, regulatory support, and market dynamics.

Advancing wireless EV charging technology

Wireless charging emerges as a convenient EV charging solution, eliminating the need for cables and plugs. It transfers power wirelessly from a ground pad to an EV's receiver plate, converting and storing energy

Industries benefiting from the EV charging boom

The surge in EV charging in the next decade will significantly impact various industries: **Charge point operators (CPOs).** These entities manage and operate charging stations, ensuring functionality and user accessibility.

Hardware manufacturers. Companies responsible for creating the physical charging units.

Installation and maintenance providers. Individuals responsible for establishing and maintaining charging infrastructures are crucial.

Electric vehicle OEMs. Makers of the electric vehicles themselves.

Component manufacturers. Key players producing AC to DC converters, DC-DC converters, and similar crucial components.

Switchgear producers. Long-standing firms making switchgears will see increased demand.

Cable, plug, and connector manufacturers. Essential for power transmission from chargers to vehicles.

Enclosure makers. Those specialising in protective casings for EV chargers, especially steel enclosures.

Display manufacturers. This sector will flourish with many EV chargers incorporating off-the-shelf LCD displays.

The growth of the EV charging system will create a ripple effect across various industries, ushering in opportunities and fostering advancements in related domains.

in the battery. Despite initial efficiency concerns, recent progress has brought wireless charging on par with traditional plug-in methods, achieving 89% to 92% efficiency for moderate power levels.

Despite its advantages, technical challenges impede broader wireless charging adoption. Power capacity remains restricted, particularly in heavy-duty applications like buses, with power levels for a single transmitter pad capped around 50 kilowatts. Additionally, high component costs stemming from low adoption rates hinder economies of scale. Consequently, wireless charging initially gains ground in premium luxury EV segments, where cost takes a backseat.

"The road to widespread wireless charging remains a complex journey, shaped by technological breakthroughs and cost optimisations. While hurdles persist, wireless charging's potential to revolutionise EV charging experiences remains a tempting prospect, gradually driving innovations that could redefine how we power our vehicles," says Shazan.

The EV ecosystem is undergoing rapid and transformative changes, both in terms of the vehicles themselves and the infrastructure that supports them. The trajectory points towards a more sustainable and efficient transportation future, from the surging numbers in EV adoption to the dynamic evolution of charging technologies. Innovations, ranging from wallbox DC chargers to the groundbreaking prospects of megawatt charging, robotic systems, off-grid solar solutions, battery swapping, and wireless charging, collectively paint a picture of a world in transition. Each of these technologies brings unique advantages and challenges, indicating the multi-faceted approach required to address the diverse needs of the global EV community. To progress on this path, it's vital to understand that technological breakthroughs don't solely drive the current momentum but also represent a collective determination to reshape transportation for a sustainable future. **EFY**

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